AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

T103/TM103
Computer Organization and Architecture

Midterm Examination (MTA)
Fall Semester 2025/2026

Date: 19/Nov./2025

Number of Exam Pages: 3 (including this cover sheet)	**Time Allowed:** 90 mins

Instructions:

- Total Marks: **30**
- This exam consists of **TWO** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Programmable Calculators** are **NOT** allowed.

CamScanner

Part 1: Multiple Choices questions [5 Marks]

Q-1: Answer all the following questions by choosing the most correct statement.

1- How many numerals are there in the octal number system?
   a. 15.
   b. 9.
   c. 8.
   d. 16.

2- Which gate produces a 1 only if all its inputs are 1 and a 0 otherwise?
   a. NAND.
   b. OR.
   c. AND.
   d. Both b and c

3- What will the equivalent of 5ns be in ms?
   a. $5 * 10^{6}$
   b. $5 * 10^{9}$
   c. $5 * 10^{-6}$
   d. $5 * 10^{-9}$

4- In the Level Hierarchy table, which layer is called the Assembly Level layer?
   a. 2.
   b. 4.
   c. 3.
   d. 6.

5- A 5T Bytes of an HDD consists of ___ bits?
   a. $5 * 10^{43}$.
   b. $5 * 2^{40}$.
   c. $5 * 2^{43}$.
   d. $5 * 10^{12}$.

PART 2: ALL QUESTIONS ARE REQUIRED [25 Marks]

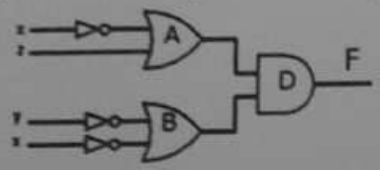
Q-2: Find the equivalent decimal number of the following Excess-16 FP representation, showing all the steps. **[2 marks]**

FP-excess 16	1	1	0	1	1	1	0	0	1	1	1	1	1

Q-3: Draw the characteristic table of the JK Flip Flop. **[2 marks]**

Q-4: Given the circuit below that implements the Boolean expression F. **[5 marks]**

a- Write the equation of F. (2 marks)



b- Using De Morgan's Law, find the complement (F') of function F. Show the solution steps. (2 marks)

c- What will be the value at the output if the input values are: (1 mark)
   x=0, y=1, z=0?

CamScanner

Q-5: Given the following Truth Table of the function F: **[5 marks]**

Inputs			F
x	y	z	
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0

a) Write the Sum of Product expression of the function F. (2 marks)
b) Draw the K-map of function F and show the groups. (2 marks)
c) Write the simplified F. (1 mark)

Q-6: Find the equivalent decimal number of 10110111 if the system uses: **[5 marks]**

a) Signed Magnitude 8-bit representation. (2 marks)
b) One's Complement 8-bit representation. (2 marks)
c) Two's Complement 8-bit representation. (1 mark)

You should show the calculations.

Q-7: Perform the following base conversions and show the calculation: **[6 marks]**

a. $(176.2)_8 = (\quad)_{10}$ [1.5 marks]
b. $(1110110.011)_2 = (\quad)_{10}$ [1.5 marks]
c. $(122.75)_{10} = (\quad)_{16}$ [1.5 marks]
d. $(1A.F)_{16} = (\quad)_8$ [1.5 marks]

End of questions

CamScanner